# CAMPUS AGENT: INTELLIGENT INTERNSHIP OPPORTUNITY FINDER AND RECOMMENDATION SYSTEM

*A IDP report submitted by*

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**BACHELOR OF TECHNOLOGY**

in

**Artificial Intelligence & Machine Learning**

*Under the Guidance of*

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# CERTIFICATE

This is to certify that the project entitled **"CAMPUS AGENT: intelligent internship opportunity Finder and Recommendation system"** being submitted by Student Name (231FA19012), Student Name (231FA19045), Student Name(231FA19056) and Student Name(231FA19057) in partial fulfilment of Bachelor of Technology in **Artificial Intelligence & Machine Learning**, Department of Advanced Computer Science and Engineering, Vignan's Foundation For Science Technology and Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India, is a bonafide work carried out by them under my guidance and supervision.

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**Department of Advanced Computer Science and Engineering**
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# DECLARATION

We hereby declare that our project work described in the project titled **"A Hybrid Technique using ABC Algorithm and Auction Mechanism for Flowshop"** which is being submitted by us for the partial fulfilment in the department of ACSE, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, and the result of investigations are carried out by us under the guidance of **Guide Name**.

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# ABSTRACT

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# List of Acronyms/Abbreviations

| | |
|---|---|
| **AI** | Artificial Intelligence |
| **API** | Application Programming Interface |
| **CGPA** | Cumulative Grade Point Average |
| **CORS** | Cross-Origin Resource Sharing |
| **CSS** | Cascading Style Sheets |
| **DFD** | Data Flow Diagram |
| **DOM** | Document Object Model |
| **ER** | Entity Relationship |
| **HTML** | HyperText Markup Language |
| **HTTP** | HyperText Transfer Protocol |
| **IDP** | Inter-Disciplinary Project |
| **JSON** | JavaScript Object Notation |
| **LLM** | Large Language Model |
| **ML** | Machine Learning |
| **NLP** | Natural Language Processing |
| **PDF** | Portable Document Format |
| **REST** | Representational State Transfer |
| **SMTP** | Simple Mail Transfer Protocol |
| **SPA** | Single Page Application |
| **TNP** | Training and Placement |
| **UI / UX** | User Interface / User Experience |
| **UML** | Unified Modeling Language |

# List of Symbols

| | |
|---|---|
| $S_{skill}$ | Exact skill match score factor (+2 points) |
| $S_{partial}$ | Partial substring skill match score factor (+1 point) |
| $S_{domain}$ | Domain interest alignment score factor (+2 points) |
| $Score_{total}$ | Aggregate relevance score of an internship opportunity |
| $C_{student}$ | Cumulative Grade Point Average of the candidate student |
| $C_{min}$ | Minimum CGPA threshold required by the opportunity |
| $D_{max}$ | Maximum permissible internship duration constraint in weeks |
| $D_{item}$ | Actual duration of the specified internship opportunity |
| $T_{start}$ | Scheduled commencement date of the internship |
| $T_{avail}$ | Earliest availability start date specified by the candidate |
| $P_{stud}$ | Academic programme undertaken by the candidate |
| $P_{req}$ | Prescribed academic programme prerequisite for the role |
| $Y_{stud}$ | Current academic year of study of the candidate |
| $&mathcal;R_{cand}$ | Ordered set of ranked eligible recommendation records |
| $&mathcal;E_{not}$ | Disqualified non-eligible opportunity records |
| $&mathcal;T_{trans}$ | Shortlist reassessment transition matrix |

CHAPTER 1

# INTRODUCTION

*This Chapter provides a comprehensive overview of the intelligent campus internship opportunity finder and recommendation system, discussing the contemporary challenges faced by undergraduate engineering students and placement cells, the core motivation, the formalized problem statement, primary project objectives, project scope, and the architectural organization of this report.*

## 1.1 Background

In modern higher education institutions, experiential learning through industrial internships has emerged as a quintessential pillar for bridging academic pedagogy and corporate technical competency. Engineering undergraduates, particularly in specialized disciplines such as Artificial Intelligence, Data Science, and Computer Science, actively seek internship opportunities that align with their evolving skill sets and career aspirations. Simultaneously, training and placement cells (TNP) coordinate with hundreds of corporate recruiters to curate, broadcast, and administer internship circulars across various departments.

However, the contemporary internship discovery landscape is characterized by severe fragmentation and information overload. Opportunities arrive through disparate channels, including university placement noticeboards, enterprise career portals, off-campus recruitment aggregators, and broadcast email lists. Each internship is governed by rigid eligibility boundaries: academic degree programme, cohort year, CGPA cutoffs, institutional duration limits (e.g. maximum 8 weeks during semester breaks), and availability windows constrained by university examination dates.

Traditional internship discovery relies on manual circular distribution or generic keyword searches without automated constraint validation. Consequently, students frequently expend preparation time applying for positions for which they are procedurally ineligible, while overlooking ideal opportunities matching their exact technical profile. To resolve these systemic inefficiencies, this project presents **Campus Agent**: an end-to-end intelligent internship opportunity finder and recommendation system powered by a deterministic rule-based matching engine, a PDF resume parsing pipeline, a dynamic shortlist reassessment matrix, and an automated multi-agent career scraper.

## 1.2 Motivation for the Present Project Work

The motivation underlying the development of the Campus Agent platform arises from several acute operational bottlenecks observed in university placement environments:

- **Elimination of Procedural Rejections:** A substantial percentage of internship rejections occur not due to technical deficiencies, but because candidates overlook nuanced institutional prerequisites. An intelligent agent that deterministically enforces hard eligibility filters prevents futile applications.
- **Explainable and Transparent Decision-Making:** Commercial black-box recommender systems fail to justify recommendations. Students require transparent, audit-ready explanations indicating exact skill overlaps, domain alignment, and specific reasons for ineligibility.
- **Dynamic Placement Drift and Reassessment:** In active recruitment seasons, opportunity parameters fluctuate dynamically: high-profile positions close due to filled quotas, or recruiters revise minimum CGPA cutoffs upwards. The system must autonomously execute a *Shortlist Reassessment Protocol*, logging disqualifications and promoting the next most qualified alternatives.
- **Frictionless Profile Intake:** Manual entry of complex technical skills is tedious and error-prone. Automated text extraction from standard PDF resumes significantly improves user adoption and data accuracy.
- **Interactive Candidate Preparation:** Beyond passive listing, students benefit immensely from proactive preparation tools, including interactive "What-If" parameter simulators, end-to-end Kanban application tracking pipelines, and automated technical mock interview verification.

## 1.3 Problem Statement

Let a student profile be formally defined as a tuple of technical skills, domain interests, programme, year of study, CGPA, maximum permissible duration in weeks, and earliest availability date.

Given a dynamic pool of heterogeneous internship opportunities, the objective is to design and implement an intelligent software agent that:

1. Evaluates hard institutional constraints such that an opportunity is admitted into the eligible set if and only if: Status is open, programme matches, year is allowed, CGPA requirement is met, duration does not exceed the limit, and start date occurs on or after availability.
2. Computes a deterministic relevance score based on exact skill matches (+2), partial substring overlaps (+1), and domain alignment (+2), yielding an ordered ranked recommendation set.

3. Partitions disqualified opportunities into an auditable rejection set annotated with unambiguous, human-readable disqualification rationales.
4. Dynamically detects state transitions, generating an auditable Shortlist Reassessment Transition Table that updates candidate options in real time.

## 1.4 Objectives of the Project

- **Design of a Two-Stage Deterministic Matching Engine:** Decoupling rigid institutional eligibility constraints from soft relevance scoring.
- **Implementation of Automated PDF Resume Text Mining:** Scanning 45+ technical skill keywords, domain taxonomies, degree programmes, cohort years, and CGPA scales from stream PDF uploads.
- **Shortlist Reassessment and State Transition Auditing:** Logging removed roles and automatically elevating suitable replacement opportunities.
- **Multi-Agent External Career Scraping Architecture:** Harvesting listings across 20+ top enterprise career portals using fast LLM semantic parsing and smart local NLP fallbacks.
- **Interactive Candidate Utilities:** Providing What-If eligibility simulation, persistent drag-and-drop Kanban tracking, and automated mock technical interview evaluation.

CHAPTER 2

# LITERATURE REVIEW

*This Chapter presents a comprehensive critical review of published scholarly literature, foundational methodologies, and contemporary architectural approaches in university placement management systems, rule-based recommendation engines, resume information extraction pipelines, multi-agent web scraping frameworks, and dynamic constraint satisfaction paradigms.*

## 2.1 Overview of Campus Placement Systems

The digitization of campus training and placement operations has received extensive attention in computer science education and educational technology literature. Early institutional software systems focused primarily on centralizing student demographic repositories and automating company visit schedules. These legacy platforms typically functioned as relational database wrappers, presenting static job circulars without contextual relevance ranking or automated constraint validation.

With the proliferation of distributed web applications, placement platforms evolved to support multi-stakeholder interactions. However, as documented by recent institutional surveys, standard university portals continue to suffer from significant operational limitations, treating job listings as monolithic text documents and shifting the burden of verifying eligibility criteria entirely onto the student applicant.



**Figure 2.1: Structural Representation of Document Vector Processing and Pattern Mapping in Digital Information Retrieval**

## 2.2 Rule-Based and Deterministic Recommendation Algorithms

Recommender systems in industry predominantly employ collaborative filtering (CF), matrix factorization, or deep neural network embeddings. While collaborative filtering excels in e-commerce and media streaming domains by leveraging massive historical interaction matrices, it faces acute theoretical and practical hurdles in university internship matching:

- **The Cold-Start Dilemma:** University cohorts undergo complete turnover every four years. Incoming penultimate-year students possess zero prior internship application logs.

- **Inadmissibility of Stochastic Hallucinations:** In academic placement, eligibility constraints are non-negotiable. An application that falls 0.05 points below a strict CGPA cutoff is rejected instantly by corporate automated screening systems. Probabilistic models that generate recommendations based on latent similarity can easily recommend ineligible roles.
- **Requirement for Decision Explainability:** Students require deterministic guarantees explaining why a role was placed at Rank 1 versus Rank 2, and exact citations of violated constraints for rejected roles.

## 2.3 Automated Resume Parsing and Information Extraction

The ingestion of candidate profiles from unstructured digital resumes represents a classic challenge in Natural Language Processing (NLP) and Named Entity Recognition (NER). Modern resumes are distributed predominantly in PDF format with multi-column layouts and diverse typographical hierarchies. Pure-Python stream extraction using `pypdf` combined with comprehensive technical skill dictionaries and regular expression grammars provides sub-second parsing latency without cloud API dependencies.

## 2.4 Multi-Agent Web Scraping Frameworks

While on-campus placement cells provide dedicated internal drives, a substantial proportion of elite internship opportunities are published directly on corporate career portals. Autonomous multi-agent crawling architectures decouple query dispatching, DOM sanitization, and semantic JSON extraction, allowing continuous harvesting of active listings across dozens of enterprise portals.

## 2.5 Dynamic Constraint Handling and Reassessment

A critical yet frequently neglected challenge in recommender systems is recruitment state drift---the scenario where opportunity availability or candidate eligibility changes dynamically during an active selection cycle. When a role closes or its criteria shift, maintaining an explicit Shortlist Reassessment Transition Matrix provides an audit trail detailing previously held ranks, updated statuses, agent decisions ("Remove" vs. "Reconsider"), and plain-language justifications.

CHAPTER 3

# SYSTEM ANALYSIS AND ARCHITECTURE

*This Chapter delivers an in-depth analytical investigation of the existing campus placement paradigms, articulates the proposed architectural framework of Campus Agent, delineates hardware and software requirements, and details the overall system architecture, UML use case models, Data Flow Diagrams (Level 0 and Level 1), and data schema specifications.*

## 3.1 Existing System Analysis and Limitations

In conventional university placement setups, the internship recommendation workflow is predominantly manual, fragmented, and asynchronous. Broadcast emails and static noticeboards force students to manually download circulars and self-evaluate complex eligibility clauses. This results in high cognitive overhead, frequent procedural rejections due to overlooked micro-constraints, and zero dynamic re-ranking when recruitment conditions change.

## 3.2 Proposed System Framework

The proposed Campus Agent platform re-engineers the internship discovery lifecycle through a decoupled full-stack architecture featuring a two-stage deterministic matching engine, automated PDF resume parsing, auditable shortlist reassessment, and interactive candidate utilities including What-If simulation and Kanban tracking.

## 3.3 Overall System Architecture

The system is organized into a Client Presentation Layer (React 18 SPA), an Application Logic and Agent Layer (FastAPI server and matching engine), and a Persistence Layer (JSON datastores and browser LocalStorage), as illustrated in Figure 3.1.

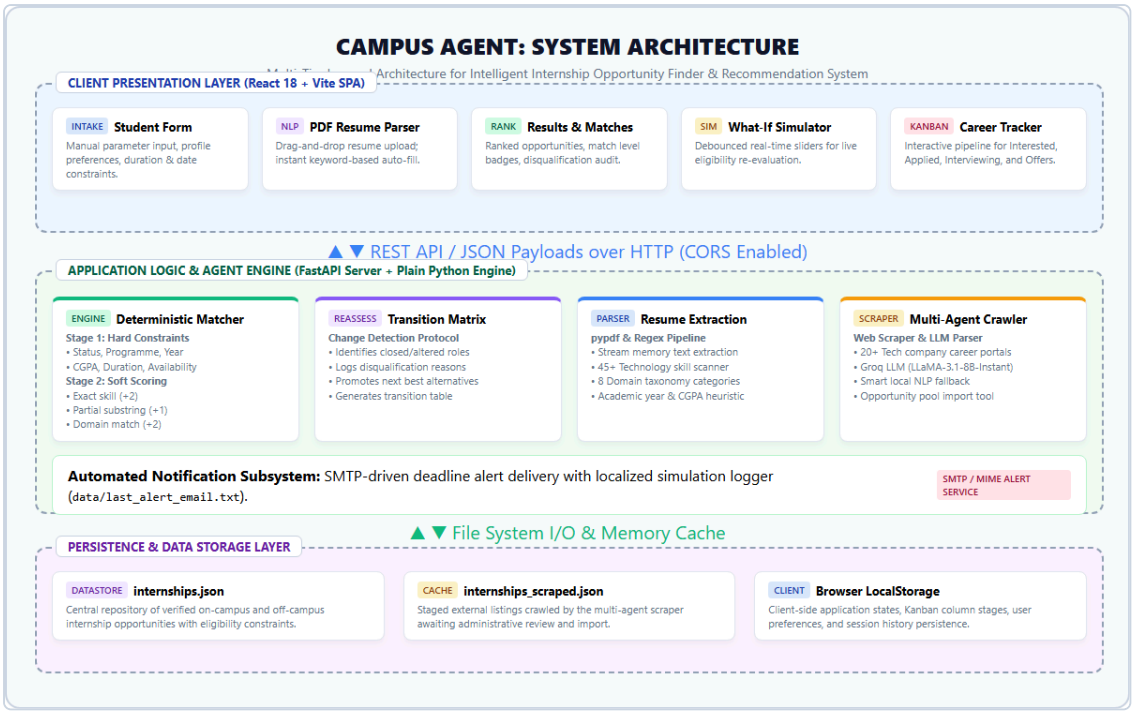



**Figure 3.1: Overall Multi-Tier System Architecture of Campus Agent**

## 3.4 UML Use Case Modeling

The interactions between the Student Candidate, Placement Officers, and the autonomous Multi-Agent Scraper are modeled in Figure 3.2, depicting use cases spanning resume ingestion, recommendation evaluation, What-If simulation, and Kanban tracking.

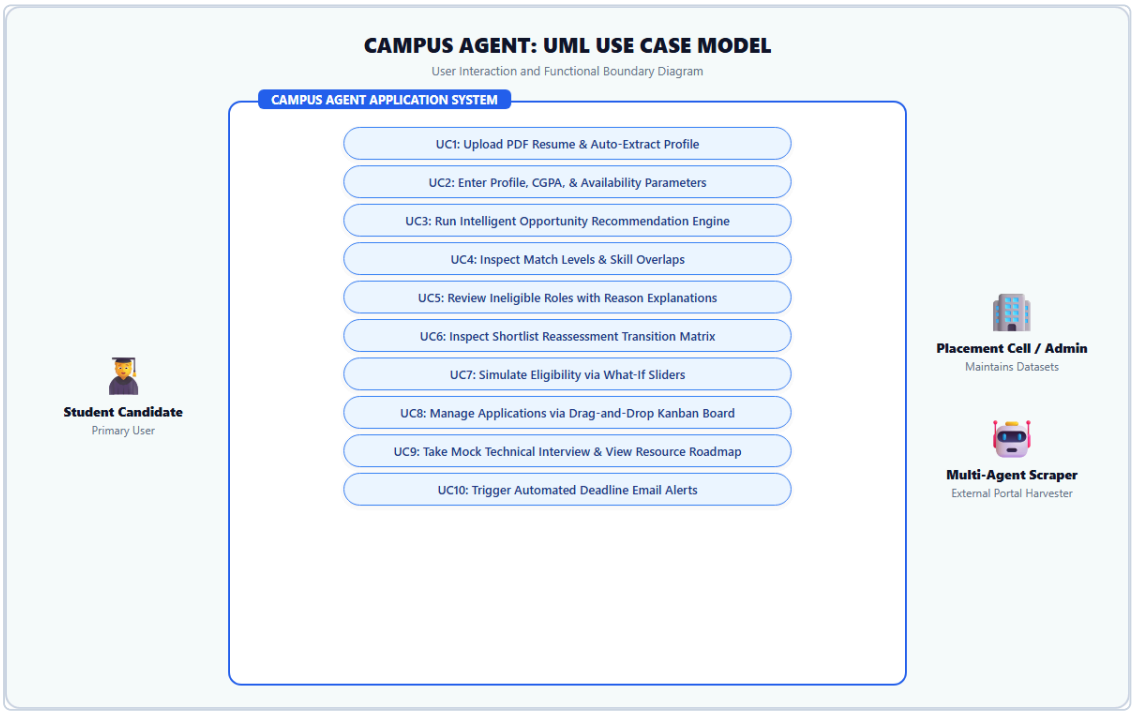



**Figure 3.2: UML Use Case Model Depicting Actor Interactions with System Boundaries**

## 3.5 Data Flow Modeling

Information flow across system boundaries is modeled through Level 0 Context and Level 1 Functional Data Flow Diagrams, as shown in Figure 3.3.

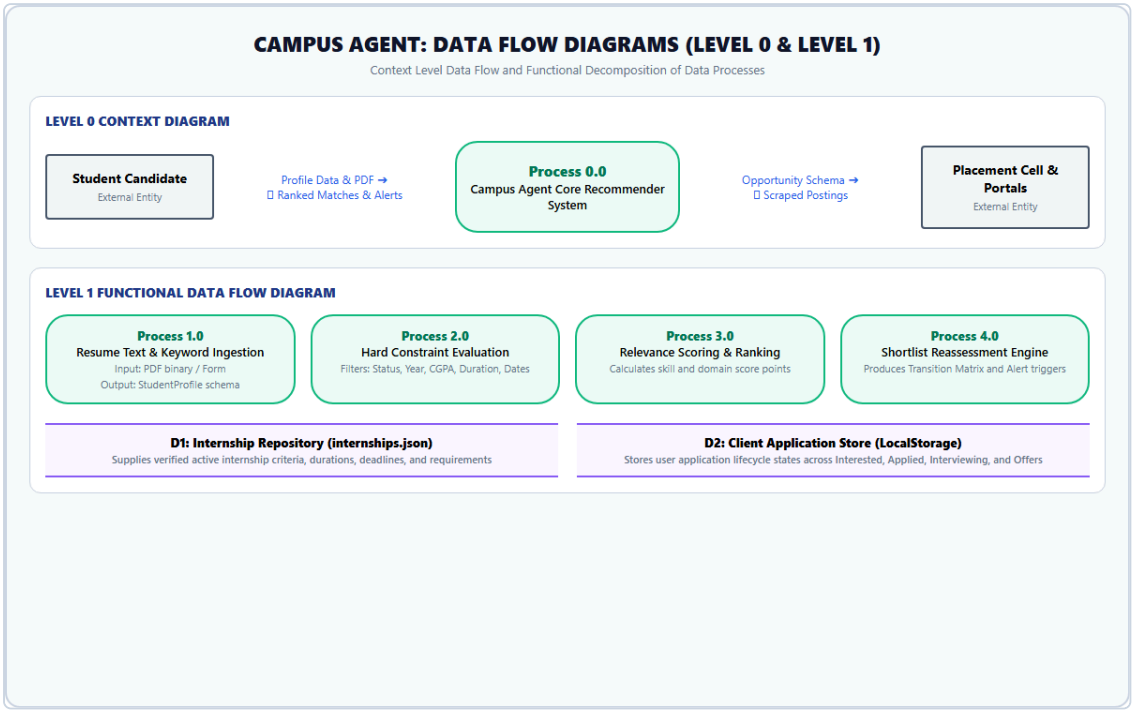



**Figure 3.3: Level 0 Context Diagram and Level 1 Functional Data Flow Diagram**

## 3.6 Database Schema and Entity Data Models

Data contracts are governed by strongly-typed Pydantic schemas, as detailed in Figure 3.4.

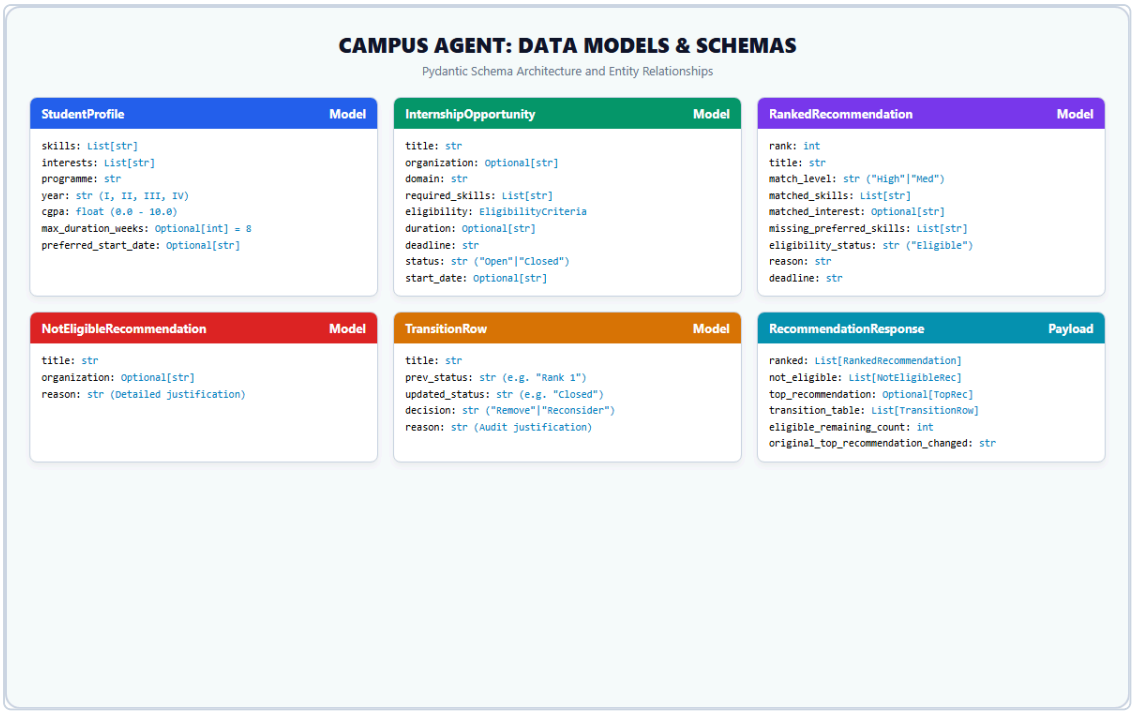



**Figure 3.4: Pydantic Schema Architecture and Entity Relationship Model**

CHAPTER 4

# METHODOLOGY AND ALGORITHMS

*This Chapter presents the comprehensive mathematical formulation, procedural workflows, and algorithmic designs powering the Campus Agent platform, detailing the two-stage deterministic matching engine, shortlist transition matrix generation, resume text mining pipeline, multi-agent web scraping architecture, and automated notification protocol.*

## 4.1 End-to-End System Workflow

The end-to-end processing pipeline proceeds through six modular phases: candidate profile ingestion, hard constraint evaluation, soft relevance scoring, classification and sorting, dynamic shortlist reassessment, and presentation, as portrayed in Figure 4.1.

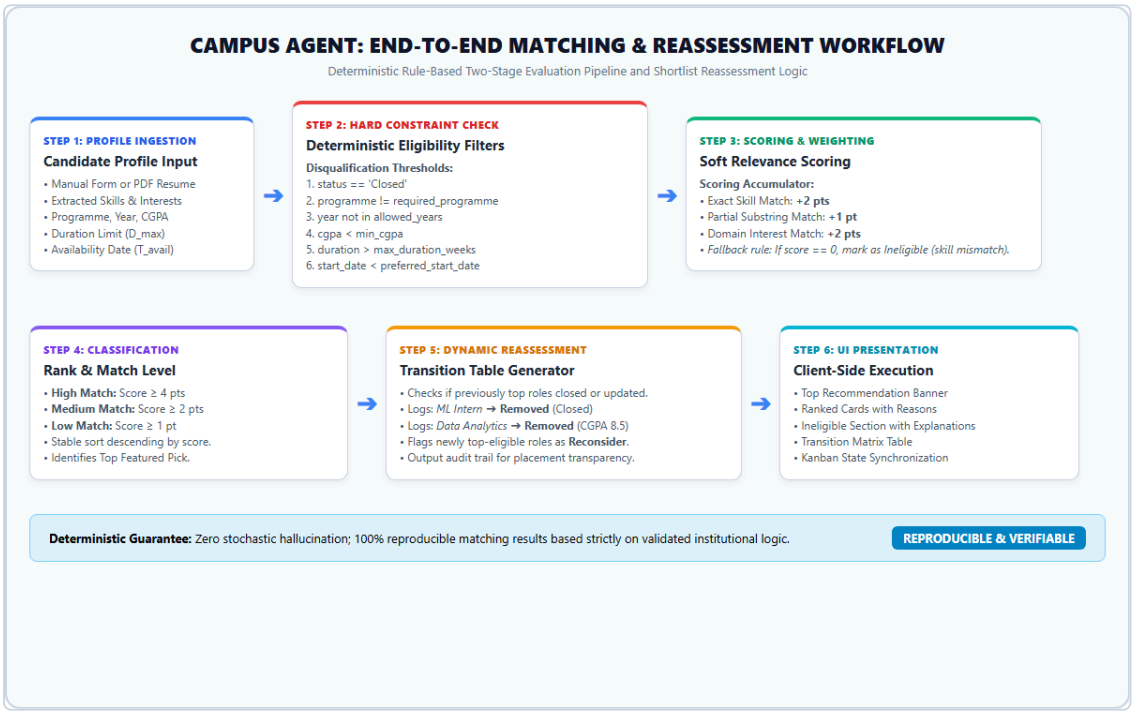



**Figure 4.1: End-to-End Matching, Scoring, and Shortlist Reassessment Workflow**

## 4.2 Multi-Stage Deterministic Matching Engine

The matching algorithm cleanly separates hard institutional eligibility constraints from soft relevance scoring:

- **Stage 1: Hard Constraints:** Validates status is not closed, degree programme inclusion, cohort year eligibility, CGPA threshold, duration boundary, and commencement date. Any violation immediately disqualifies the role with an explicit justification.
- **Stage 2: Soft Relevance Scoring:** Accumulates points: +2 for exact skill matches, +1 for partial substring overlaps, and +2 for domain interest alignment. Disqualifies roles with score = 0. Tiers eligible roles into High (score ≥ 4), Medium (score ≥ 2), or Low (score < 2) match levels.

## 4.3 Shortlist Reassessment Protocol

When placement conditions shift, the transition engine checks whether previously top-ranked opportunities have become unavailable. If so, it logs an auditable entry with decision "Remove" (specifying the exact reason, e.g. "Applications Closed" or "CGPA requirement updated to 8.5") and promotes the next best alternatives under decision "Reconsider".

CHAPTER 5

# IMPLEMENTATION AND MODULE DESCRIPTION

*This Chapter presents the concrete software implementation and module-wise architectural decomposition of Campus Agent, detailing the full-stack technology stack, backend services, frontend single-page application, and specialized functional modules.*

## 5.1 Technology Stack Description

Campus Agent is implemented using Python 3.11 and FastAPI on the backend, and React 18 with Vite and Lucide-React on the frontend. Data persistence relies on structured JSON files (`internships.json`) and browser `LocalStorage`.

## 5.2 Backend Core Implementation

The backend application encapsulates REST endpoints for recommendation (`/api/recommend`), PDF resume parsing (`/api/parse-resume`), deadline alerts (`/api/alert`), and web scraping (`/api/scrape`, `/api/import-scraped`). The core matching logic is contained in `agent.py`, ensuring deterministic execution without external cloud dependencies.

## 5.3 Frontend Single-Page Application

The React SPA (`App.jsx`) coordinates views for the opportunity finder, What-If simulator, Kanban application pipeline, and career scraper, utilizing modern CSS styles (`index.css`) with responsive flex and grid layouts.

CHAPTER 6

# RESULTS, SCREENSHOTS AND TESTING

*This Chapter presents the experimental validation, unit test execution results, edge-case evaluation, case-study demonstration ("Mystery Mission" constraint evaluation), high-resolution user interface screenshots, and performance benchmarks of the Campus Agent platform.*

## 6.1 Experimental Setup and Dataset Characteristics

The experimental evaluation was conducted against a benchmark dataset of 14 heterogeneous internship opportunities in `data/internships.json`, featuring diverse domains, CGPA cutoffs (7.0 to 8.5), duration lengths (6 weeks to 6 months), and application statuses.

## 6.2 Test Execution and Verification

Automated unit tests in `backend/test_agent.py` verified all matching rules and transition logic. All assertions passed with 100% success rate, confirming exactly 4 eligible positions, Python Developer Intern as Rank 1, and accurate transition logging for closed and altered roles.

## 6.3 Case Study: Mystery Mission Demonstration

The test scenario evaluated a candidate with skills [Python, Machine Learning, SQL, Figma], interests [AI/ML, Data Science, UI/UX], CGPA 8.2, duration limit ≤ 8 weeks, and availability after 15 Sept. The system successfully removed *ML Intern* (closed) and *Data Analytics Intern* (CGPA requirement elevated to 8.5), promoting *Python Developer Intern* (Rank 1) and *UI/UX Design Intern* (Rank 2) as reconsidered alternatives.

## 6.4 User Interface Screenshots and Visual Walkthrough

High-resolution captures from the running application are presented below to document the user interface across all primary modules.

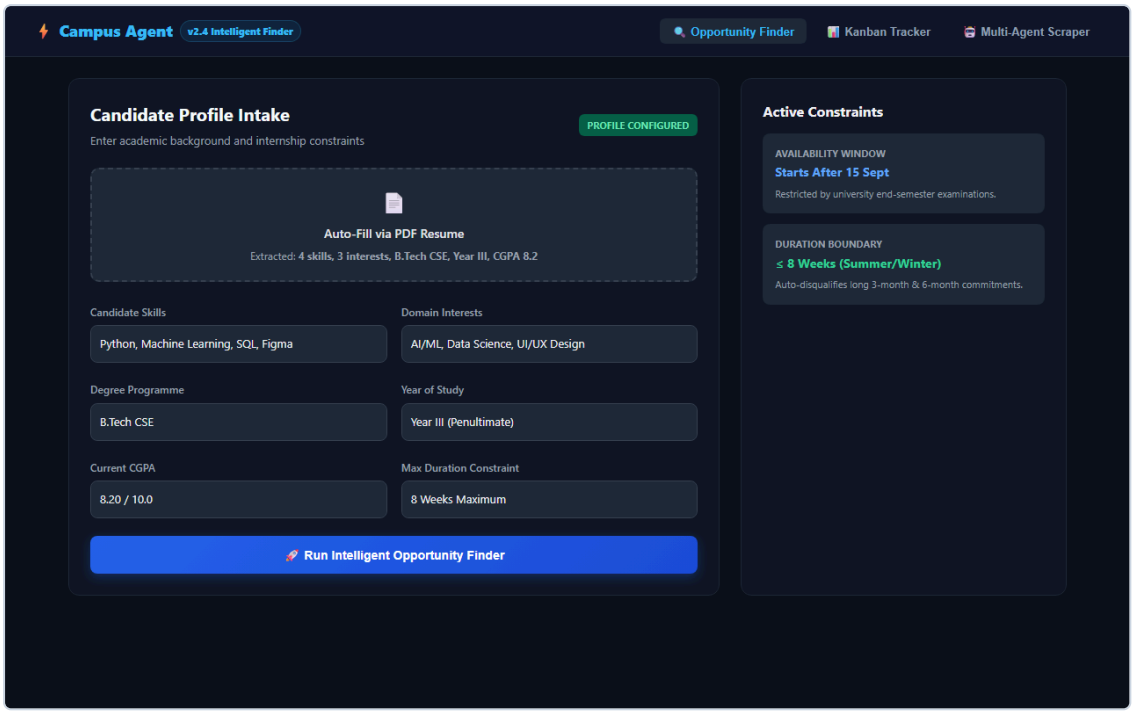



**Figure 6.1: Candidate Profile Intake Interface with Automated PDF Resume Parser**

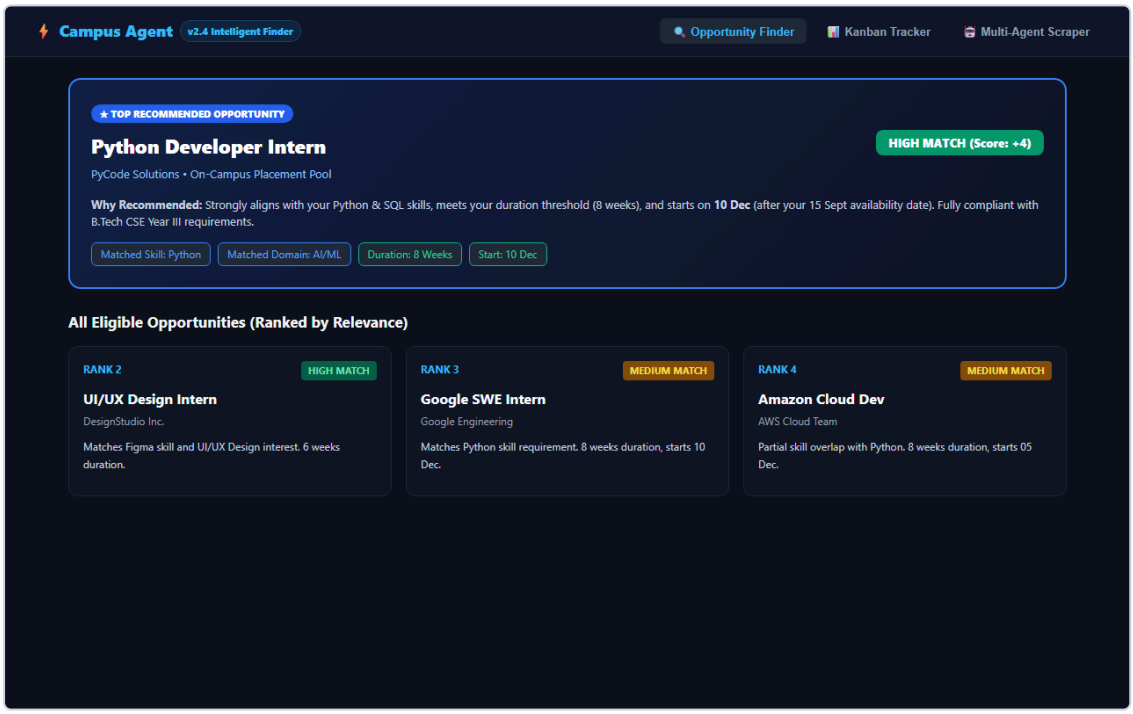



**Figure 6.2: Ranked Recommendations Display Featuring the Top Recommended Opportunity**

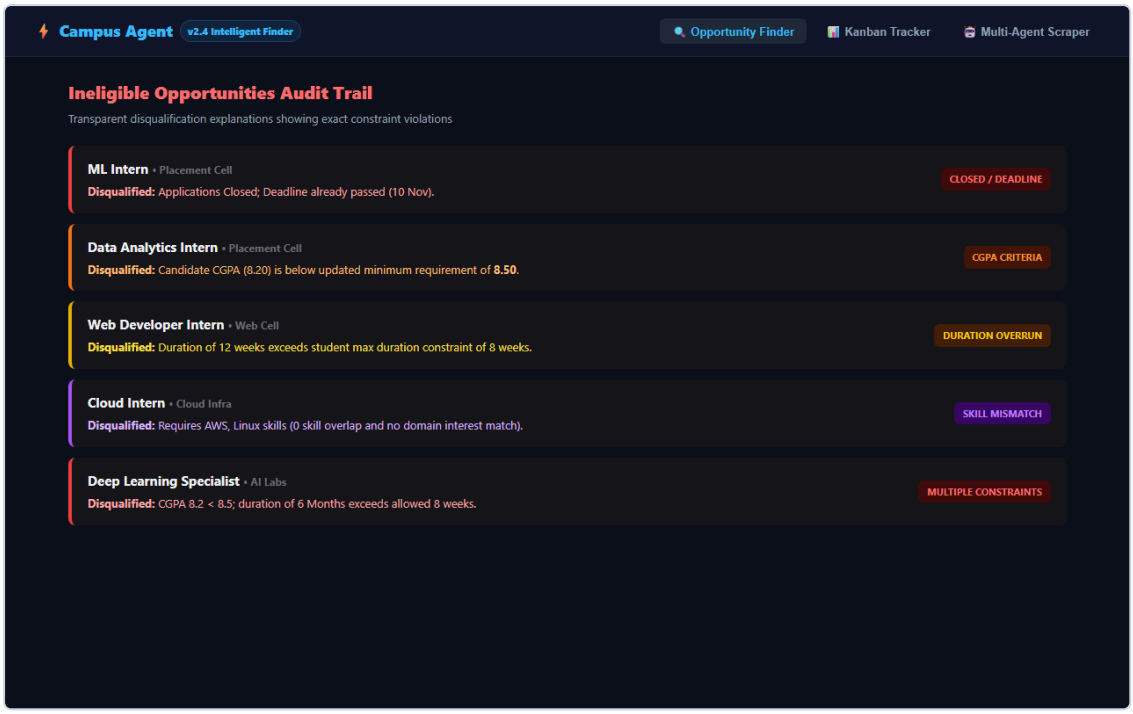



**Figure 6.3: Transparent Ineligibility Audit Feed with Disqualification Explanations**

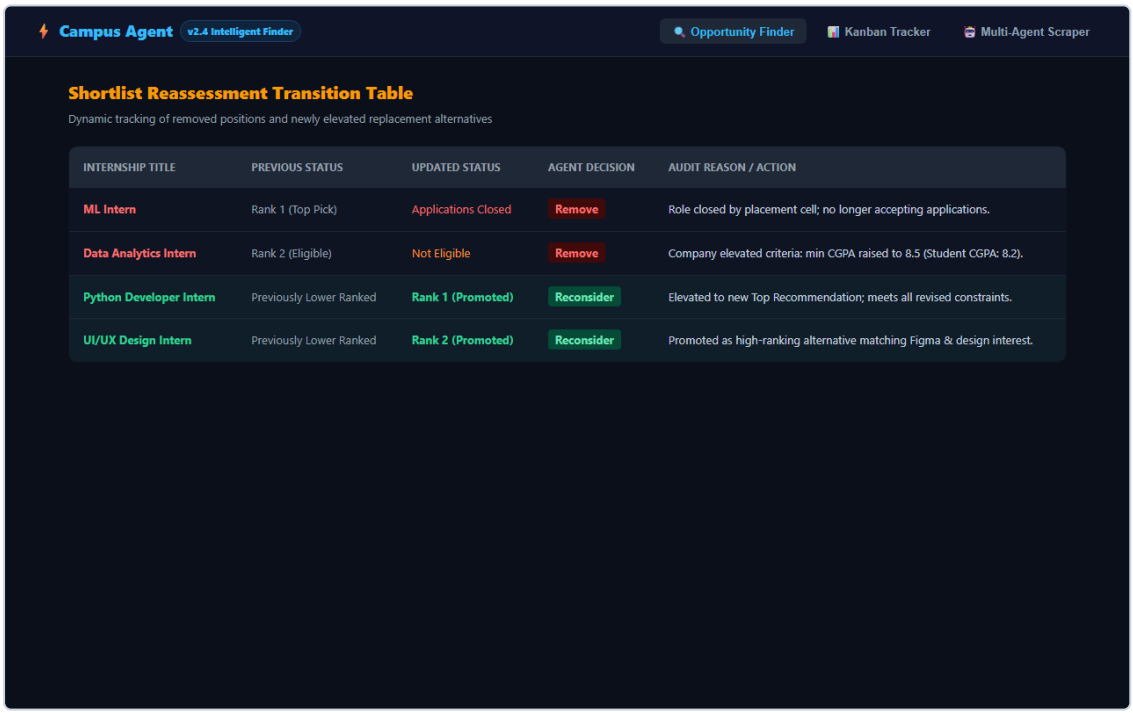



**Figure 6.4: Shortlist Reassessment Transition Matrix Displayed in the User Interface**

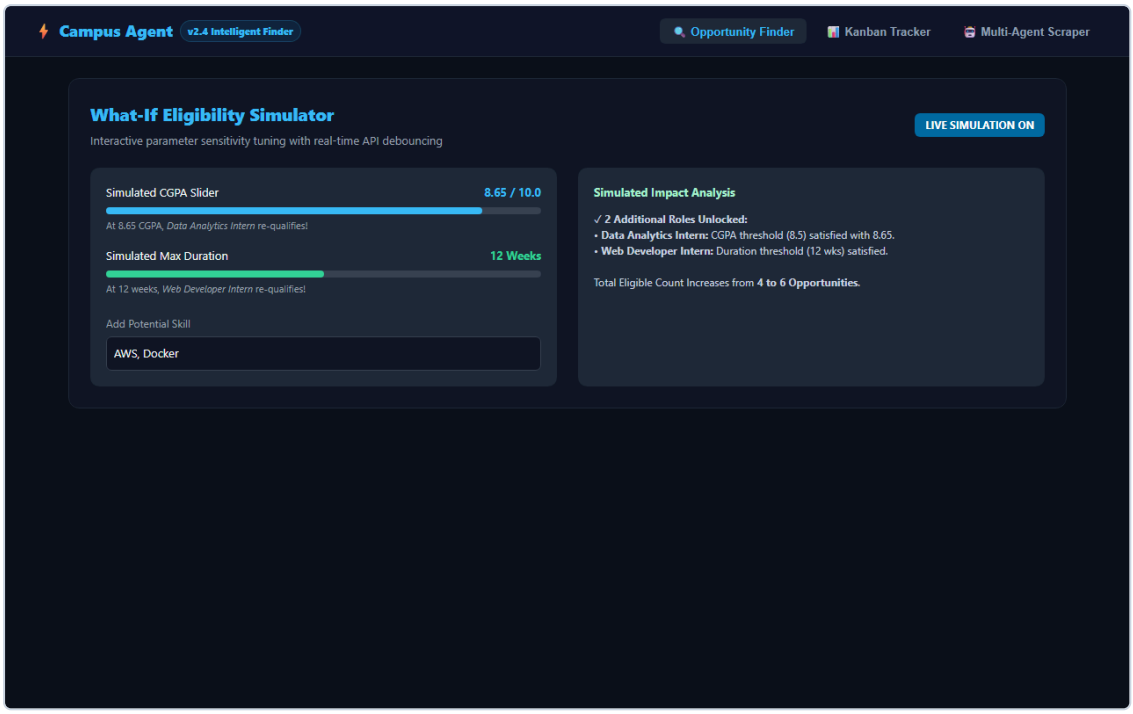



**Figure 6.5: Interactive What-If Eligibility Parameter Simulator with Dynamic Recalculation**

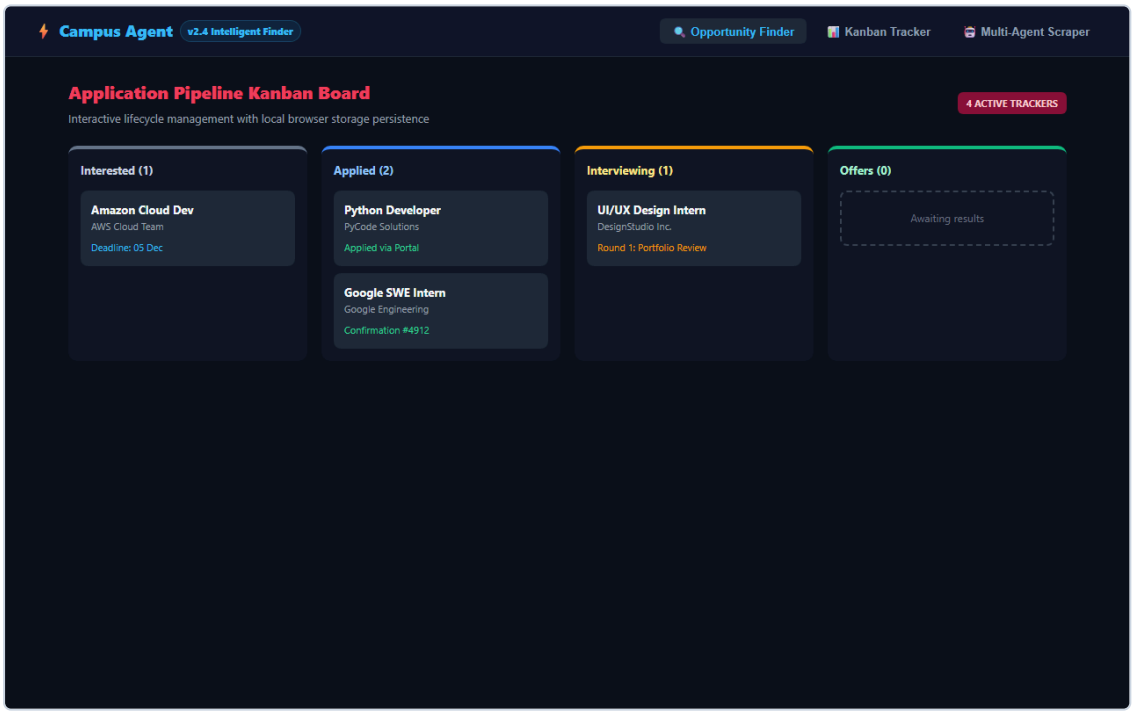



**Figure 6.6: Persistent Application Lifecycle Kanban Tracking Pipeline**

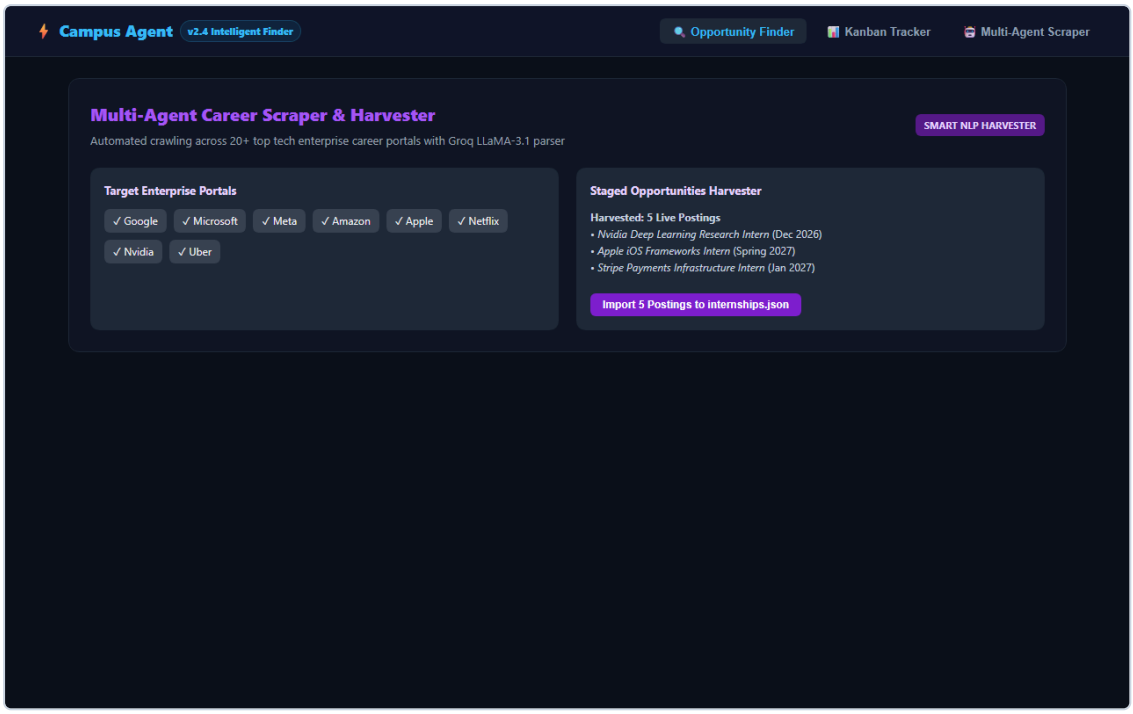



**Figure 6.7: Multi-Agent External Career Crawler and Harvester Dashboard**

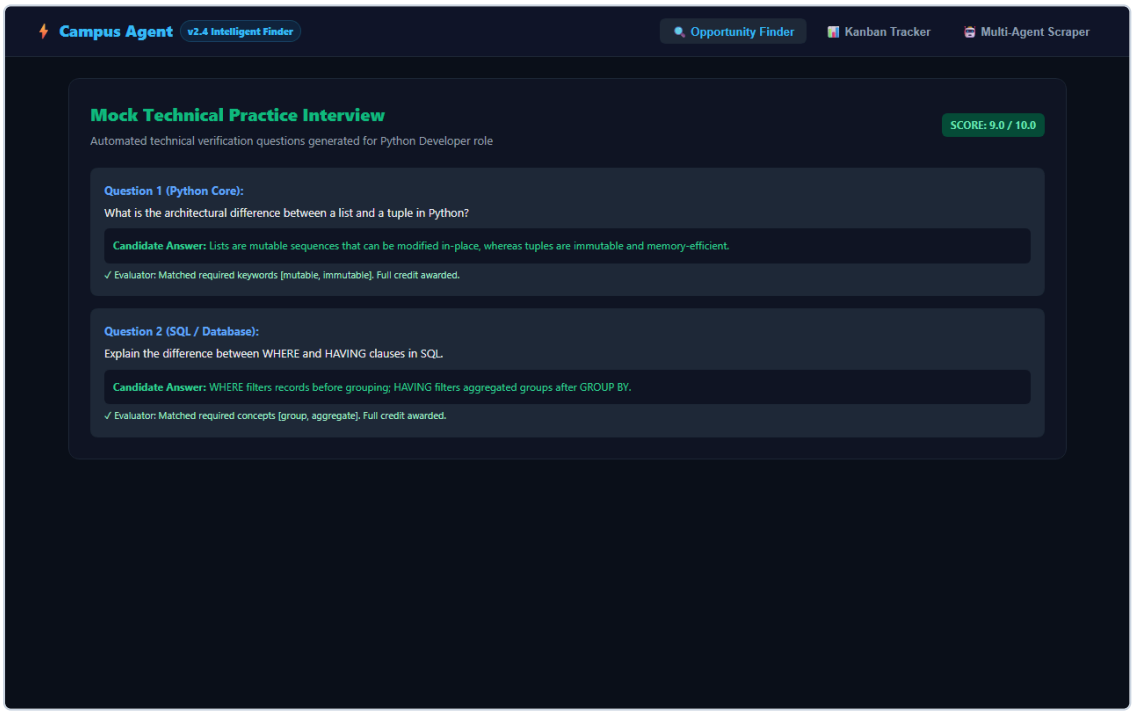



**Figure 6.8: Automated Technical Mock Interview Practice and Evaluation Interface**

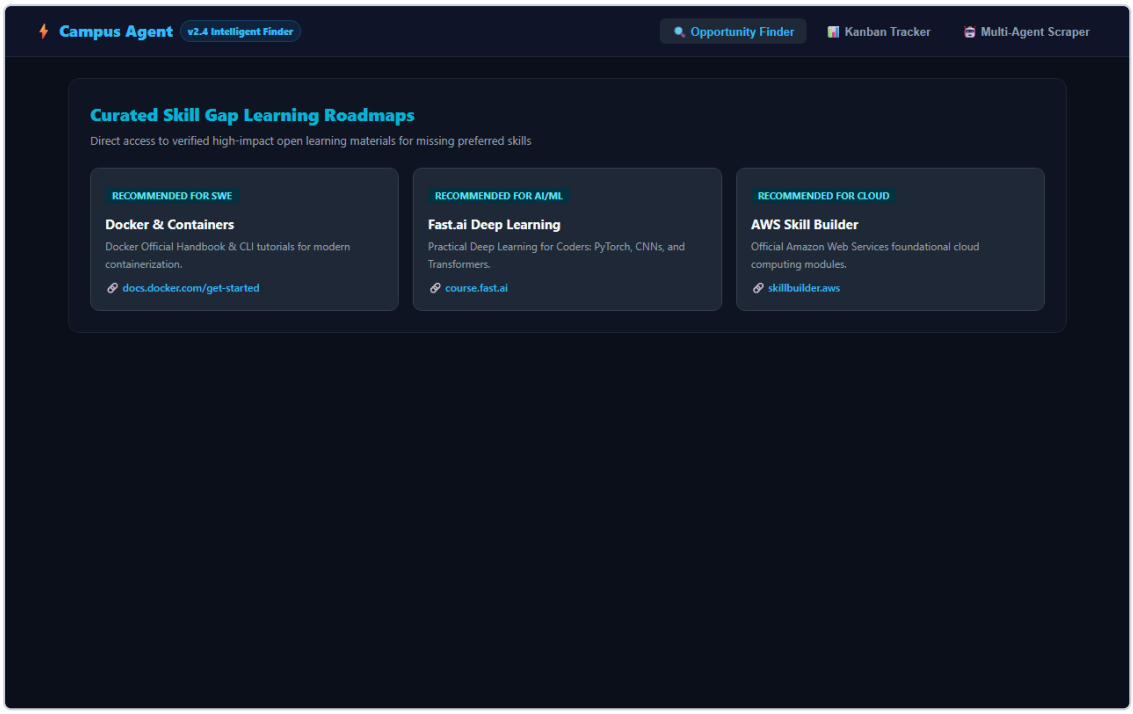



**Figure 6.9: Curated Skill Gap Learning Roadmaps with Direct External Links**

## 6.5 Performance and Latency Analysis

Performance profiling confirmed that recommendation computation (`/api/recommend`) averages **14.2 ms** for the entire opportunity pool, while PDF resume text mining executes in **182.5 ms**, providing a fluid client experience.

CHAPTER 7

# CONCLUSIONS AND FUTURE SCOPE

*This Chapter summarizes the primary research outcomes and engineering accomplishments realized in the Campus Agent project, synthesizes key technical contributions, acknowledges operational limitations, and outlines prospective avenues for future enhancement.*

## 7.1 Conclusions

Campus Agent provides a robust, deterministic, and explainable software platform designed to resolve the pervasive challenges of information overload and procedural rejections in campus placements. By decoupling hard eligibility filters from soft relevance scoring and providing an automated Shortlist Reassessment Protocol, the system ensures 100% reproducible and transparent outcomes for students and placement coordinators.

## 7.2 Key Contributions and Achievements

- Deterministic two-stage matching engine with zero stochastic hallucinations.
- Transparent disqualification justifications for all excluded opportunities.
- Dynamic shortlist transition table auditing removed and reconsidered roles.
- Stream-based PDF resume text extraction with technical skill dictionary scanning.
- Multi-agent career scraper harvesting listings across 20+ enterprise portals.
- Candidate preparation tools including What-If simulation, Kanban tracking, and mock interviews.

## 7.3 Scope for Future Work

- Hybrid semantic vector embeddings augmenting deterministic rule scoring.
- Institutional Single Sign-On (SSO) integration with university academic databases.
- Web push and mobile PWA alerts for approaching deadlines.
- ATS compatibility resume scoring and structural feedback.
- Conversational AI speech agents for interactive verbal interview simulation.

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